

Gen-ai ~ **Assignment - 2**

(Project - 1)

Memory-Based Chatbot Using ChromaDB and Groq API

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Project Overview

Explain:

- The chatbot remembers important user information.
- ChromaDB stores memories as vectors.
- Sentence Transformers generate embeddings.
- Groq API generates responses.
- Similarity search retrieves relevant memories.

Concepts Implemented

- **Embeddings**

The chatbot uses Sentence Transformers to convert user messages into numerical vector representations (embeddings). These embeddings capture the semantic meaning of the text, enabling the system to understand context beyond exact keyword matching.

- **Chunking**

Long user inputs are divided into smaller chunks before storage. This improves retrieval efficiency and ensures that relevant information can be located accurately when needed.

- **Vector Storage**

Generated embeddings are stored as vectors, allowing the chatbot to maintain long-term memory and efficiently search through previously stored information.

- **Similarity Search**

When a user asks a question, the query is converted into an embedding and compared against stored vectors. The chatbot retrieves memories that are semantically similar to the user's query.

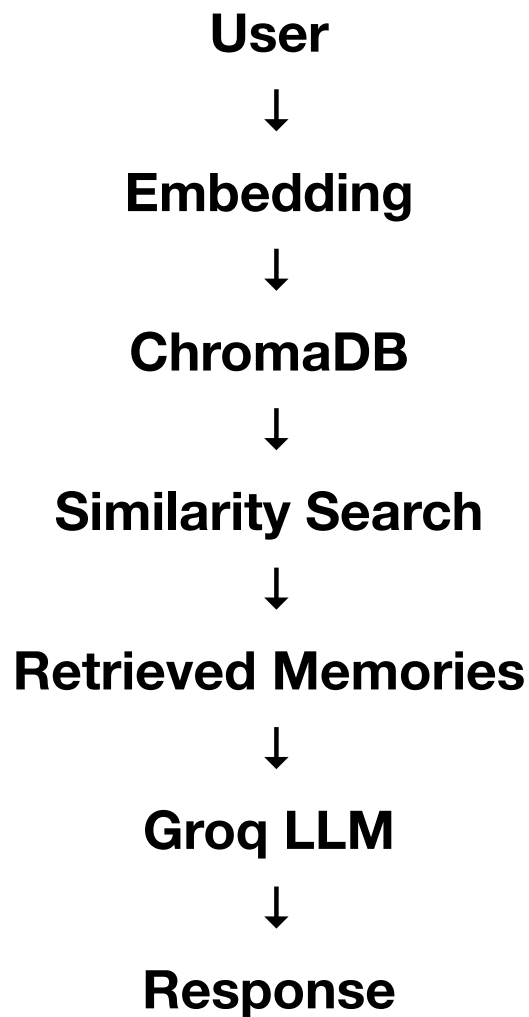
- **ChromaDB**

ChromaDB serves as the vector database for the project. It stores embeddings and performs fast similarity searches to identify relevant memories from previous interactions.

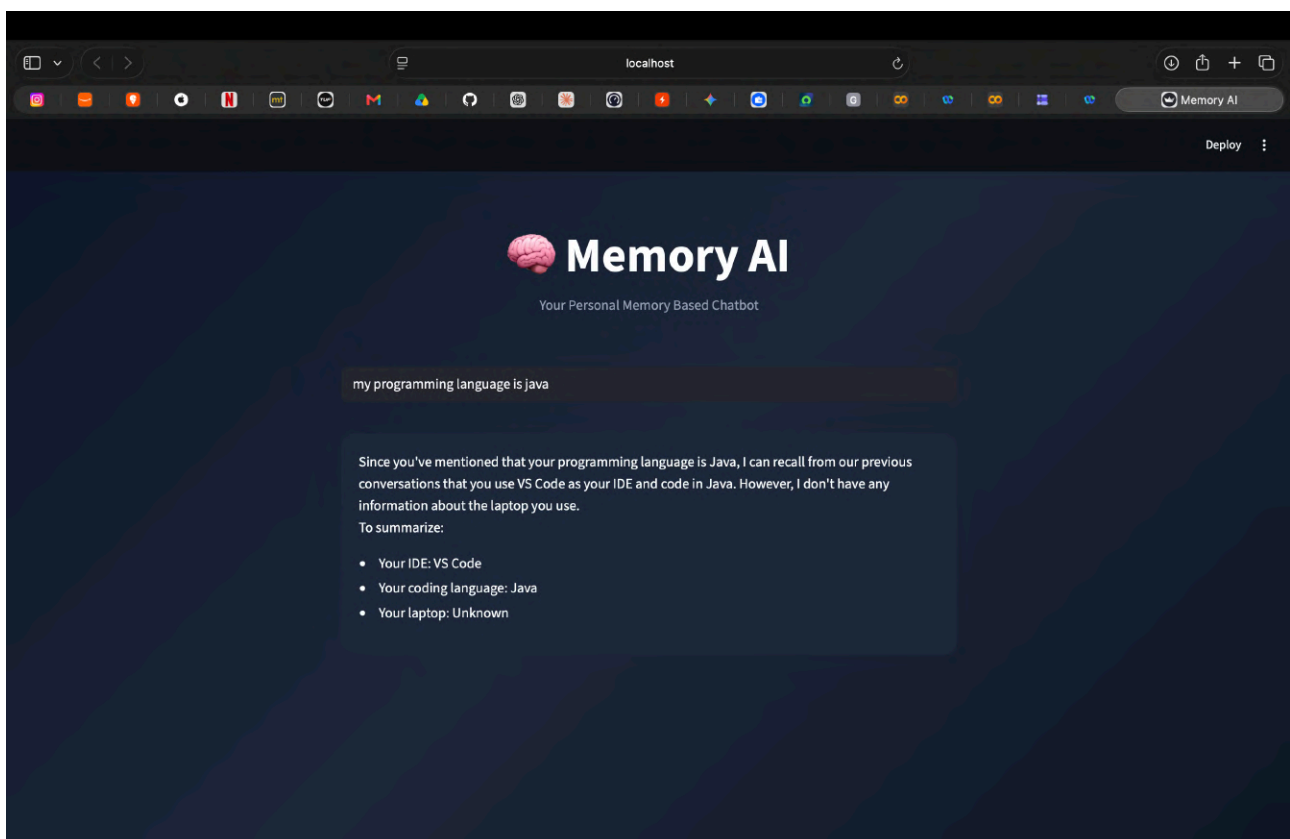
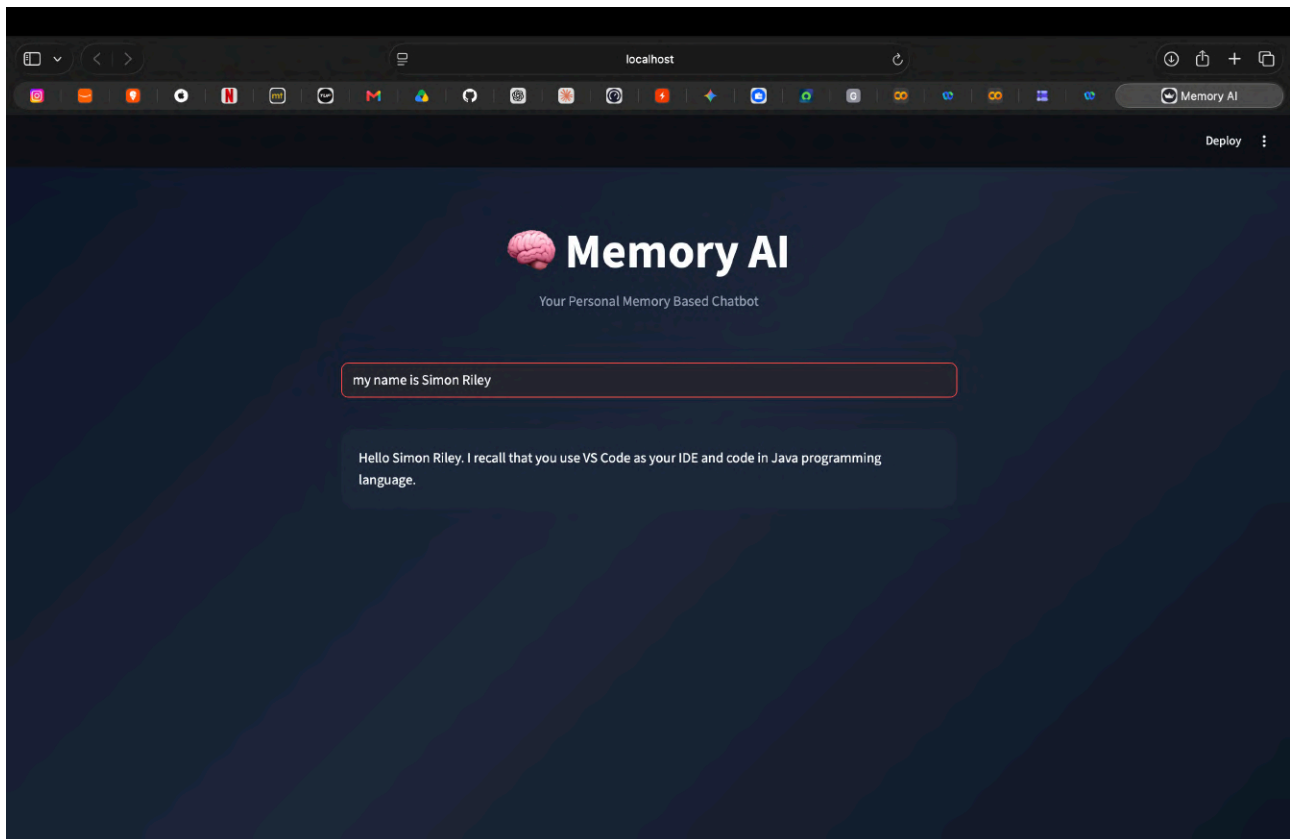
- **Groq API**

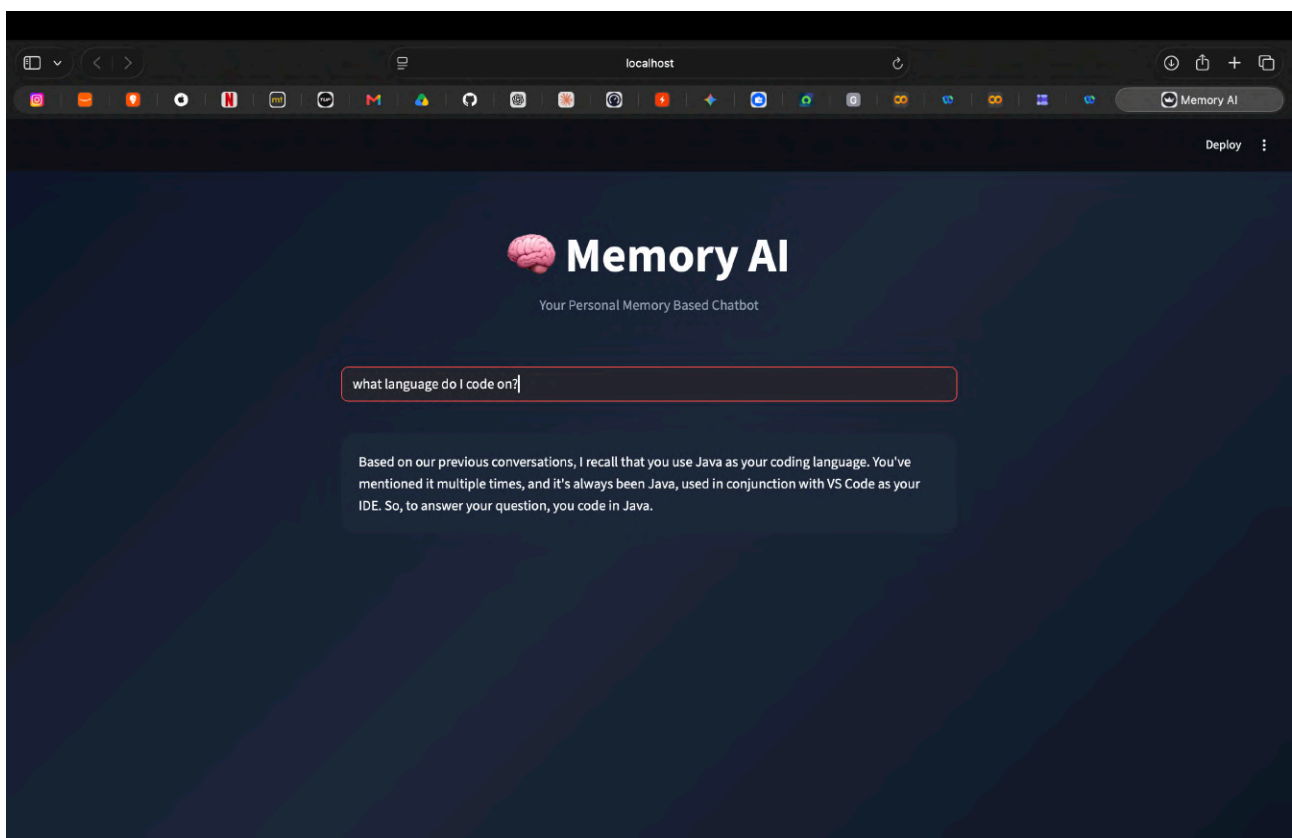
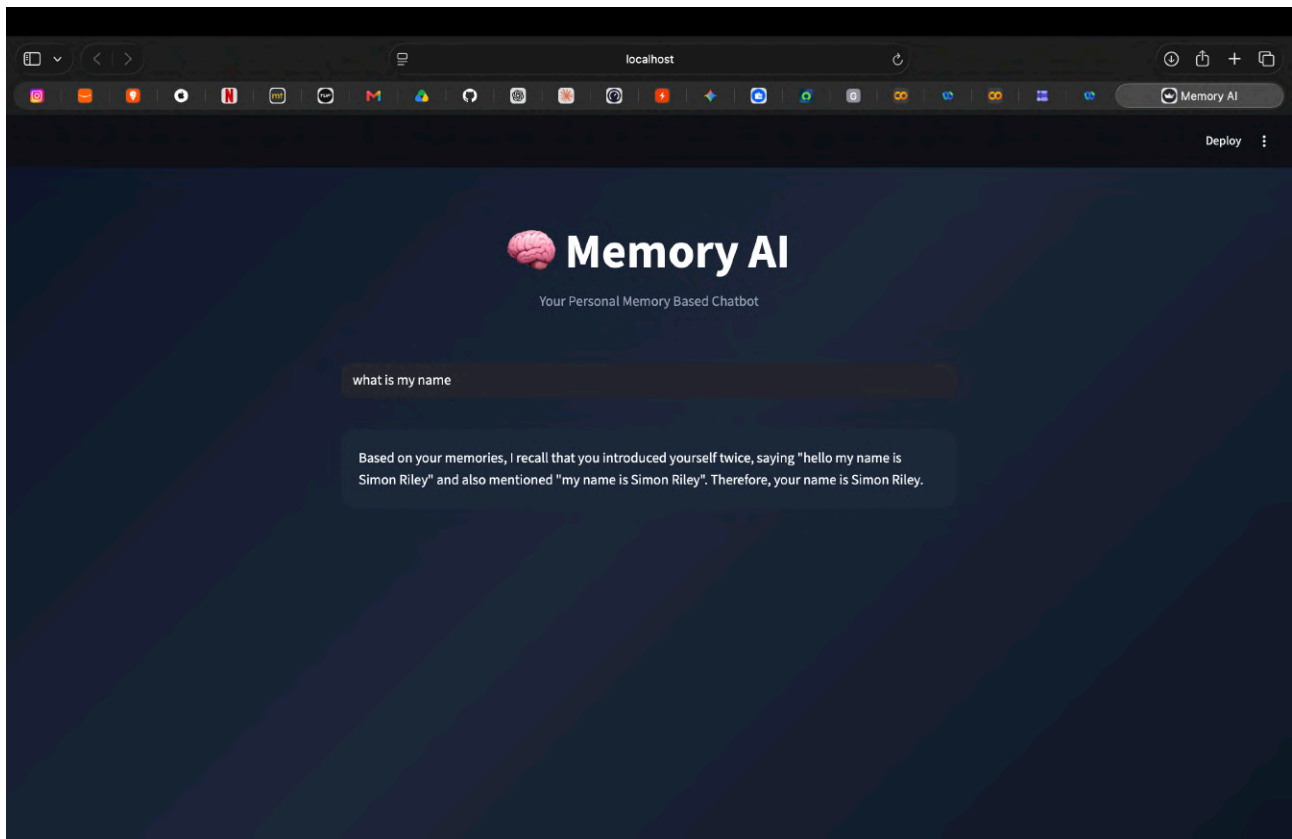
The Groq API powers the Large Language Model (LLM) used in the chatbot. Retrieved memories are provided as context to the model, enabling personalised and context-aware responses.

Architecture Diagram



GitHub : <https://github.com/CrwazzyCupcakes/Gen-ai-Summer-Internship->





Gen-ai ~ **Assignment - 2**

(Project - 2)

**AI PDF Assistant using
ChromaDB and Groq API**

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Project Overview

Develop an AI-powered chatbot that can read PDF documents, store their contents in a vector database, and answer user questions based on the uploaded document.

Concepts Implemented

- **Embeddings**

The chatbot uses Sentence Transformers to convert user messages into numerical vector representations (embeddings). These embeddings capture the semantic meaning of the text, enabling the system to understand context beyond exact keyword matching.

- **Chunking**

Long user inputs are divided into smaller chunks before storage. This improves retrieval efficiency and ensures that relevant information can be located accurately when needed.

- **Vector Storage**

Generated embeddings are stored as vectors, allowing the chatbot to maintain long-term memory and efficiently search through previously stored information.

- **Similarity Search**

When a user asks a question, the query is converted into an embedding and compared against stored vectors. The chatbot retrieves memories that are semantically similar to the user's query.

- **ChromaDB**

ChromaDB serves as the vector database for the project. It stores embeddings and performs fast similarity searches to identify relevant memories from previous interactions.

- **Groq API**

The Groq API powers the Large Language Model (LLM) used in the chatbot. Retrieved memories are provided as context to the model, enabling personalised and context-aware responses.

Architecture Diagram

